

Project Proposal: Decentralized Identity and Access Management (IAM) dApp for Secure Medical Records

Team Name: Group 21

Team Members: Aswath Sridhar, Ben Stewart Silas Sargunam, Devesh Valluru, Pooja Rani Shivanand Naik, Reethika Gogireddy

Problem Statement:

Traditional healthcare systems rely on centralized databases to store patient records, which are vulnerable to security breaches, unauthorized access, and loss of data integrity. Patients have limited control over their information, making it difficult to ensure privacy and selective data sharing across hospitals and clinics. The lack of interoperability also hinders timely access to medical data, especially during emergencies.

Motivation:

The growing importance of digital health records and the increasing number of cyberattacks on centralized medical databases highlight the need for a decentralized, secure, and patient-controlled system. Blockchain technology can enable transparency, integrity, and autonomy in healthcare data management, allowing patients to control access to their medical information.

Literature Review:

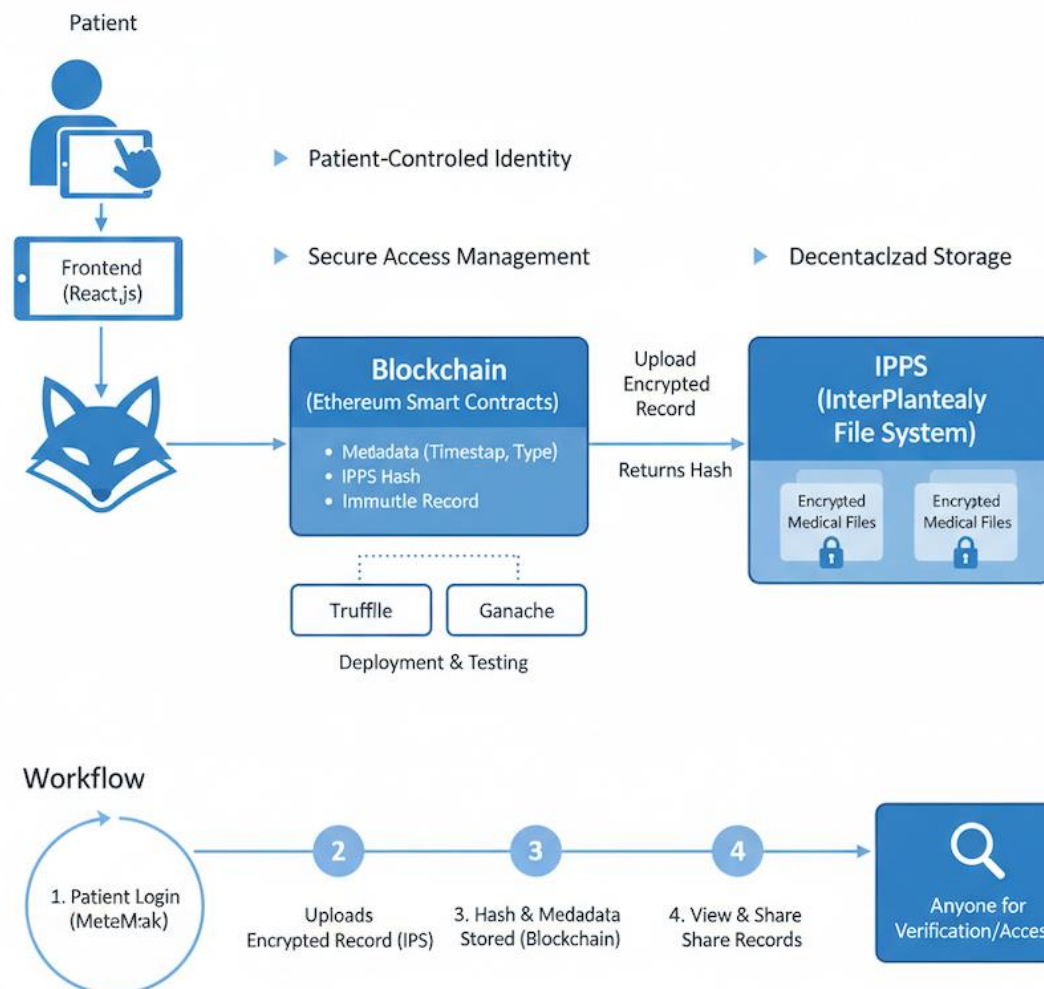
1. MedRec (MIT, 2016) – Introduced blockchain for medical data management using smart contracts on Ethereum.
2. Hyperledger in Healthcare (IBM, 2018) – Demonstrated permissioned blockchain frameworks for data sharing across institutions.
3. Recent studies (2020–2023) – Combine blockchain with IPFS for decentralized storage to achieve scalability and enhanced privacy.

Most prior systems depend on multiple user roles (doctor, hospital, and admin). Our approach simplifies implementation by focusing on a *single patient-controlled identity model* using blockchain and IPFS for secure access management.

Proposed Architecture:

- **Frontend (React.js):** User interface for patients to register (via MetaMask), upload records, and view stored data.

- **Blockchain (Ethereum Smart Contracts):** Stores the metadata and IPFS hash of each encrypted file, ensuring immutability and integrity.
- **IPFS (InterPlanetary File System):** Stores encrypted medical files off-chain, reducing blockchain overhead.
- **MetaMask + Truffle + Ganache:** Provide authentication, deployment, and testing on a local blockchain network.



Workflow:

1. The patient logs in using MetaMask (wallet = digital identity).
2. Uploads encrypted medical record → stored in IPFS → returns a hash.
3. The IPFS hash and metadata (timestamp, file type) are stored on the blockchain.

4. The patient can view all their recorded hashes and share them with anyone for verification or access.

Chosen Platform:

- **Blockchain Network:** Ethereum (Ganache test network)
- **Smart Contracts:** Solidity
- **Frontend:** React.js + MetaMask
- **Storage:** IPFS
- **Development Tools:** Truffle Suite, Node.js

References:

1. Ekblaw, A. et al. MedRec: Blockchain for Medical Data Access and Permission Management, MIT, 2016.
2. IBM Research. Blockchain in Healthcare: Improving Transparency and Trust, 2018.
3. Sharma, T. & Jindal, R. Secure Medical Data Sharing using Blockchain and IPFS, IEEE Access, 2022.

Team Contract

Team Members	Role
Aswath Sridhar	Solidity Developer 1, Documentation
Ben Stewart Silas Sargunam	Solidity Developer 2, Documentation
Pooja Rani Shivanand Naik	Backend Developer 1, Testing
Devesh Valluru	Backend Developer 2, Testing
Reethika Reddy Gogireddy	Frontend Developer, Documentation

Expectations:

- Communicate actively via Teams/Slack.
- Commit code to GitHub with peer reviews.
- Attend two weekly sync meetings.
- Complete assigned tasks before deadlines; support teammates during milestone weeks.

Meeting Frequency:

Twice per week (additional ad-hoc sessions near submissions). Once during Thursday during class hours. And next Mondays online to check progress.